Cloud-Based Active Directory and User Management.

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Project Objective

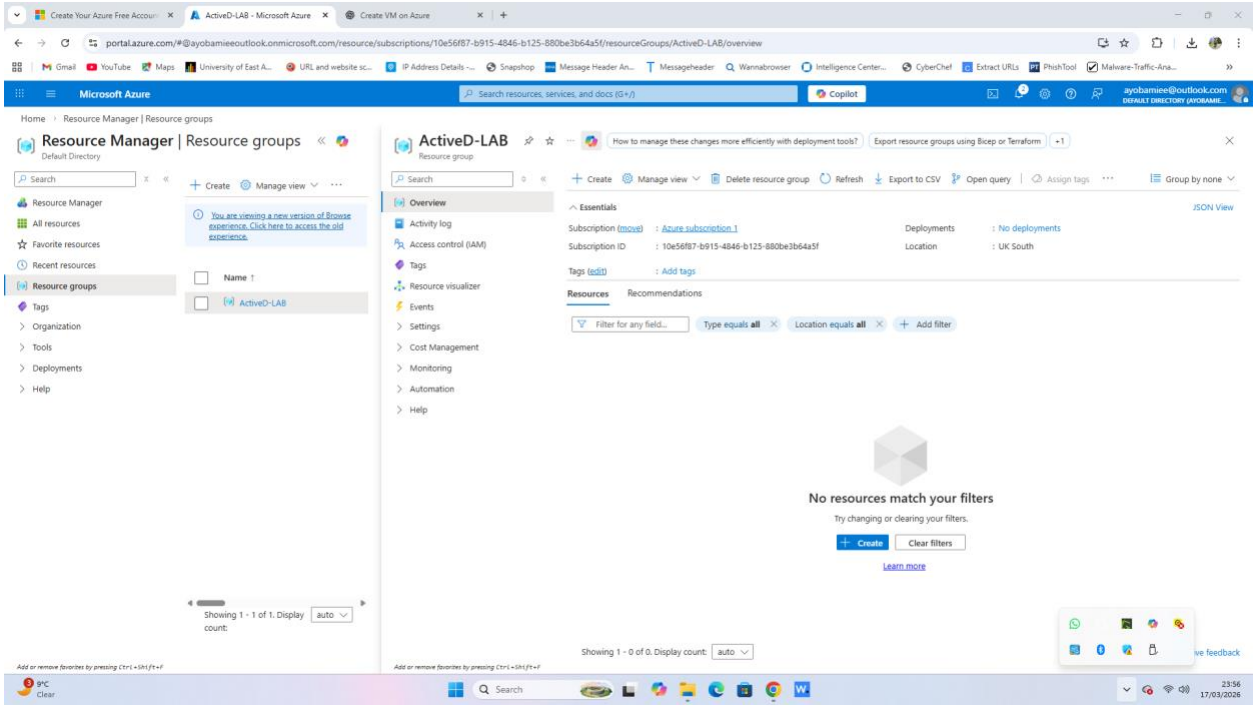
This project focuses on the design, deployment, and administration of a cloud-based Active Directory environment within Microsoft Azure, with the objective of simulating a real-world enterprise identity infrastructure. The lab environment is structured using Azure IaaS components, including a dedicated resource group and virtual network, to replicate secure network segmentation and communication.

Two Windows Server 2022 virtual machines were provisioned to establish a functional domain environment: a domain controller configured with Active Directory Domain Services (AD DS) and a domain-joined client workstation to emulate typical user activity. Core configurations include domain creation, client domain integration, and user account management.

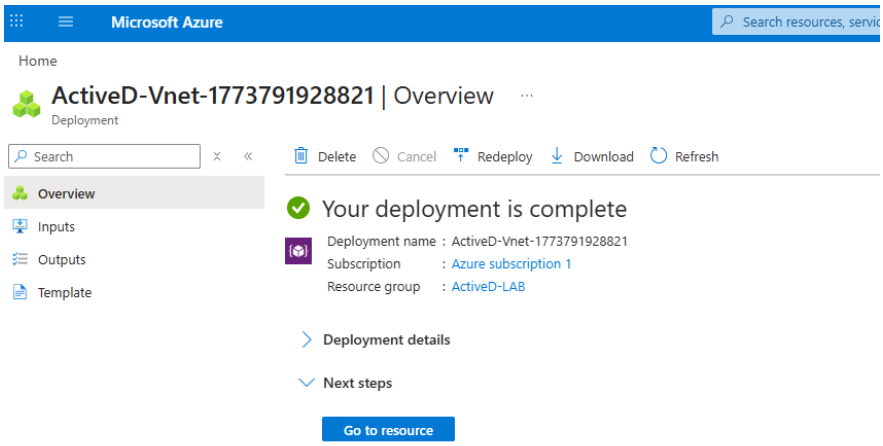
From a security operations perspective, this project is designed to build foundational SOC analyst skills by enabling visibility into authentication processes, identity management, and administrative actions. It provides a practical framework for understanding how centralized identity systems generate security-relevant events that can be monitored, analysed, and leveraged for threat detection in enterprise environments.

Microsoft Account Set up and Deployment:

- Create a Microsoft account and access the azure portal.
- Create a Resource group on the Azure portal named **(ActiveD-LAB)**

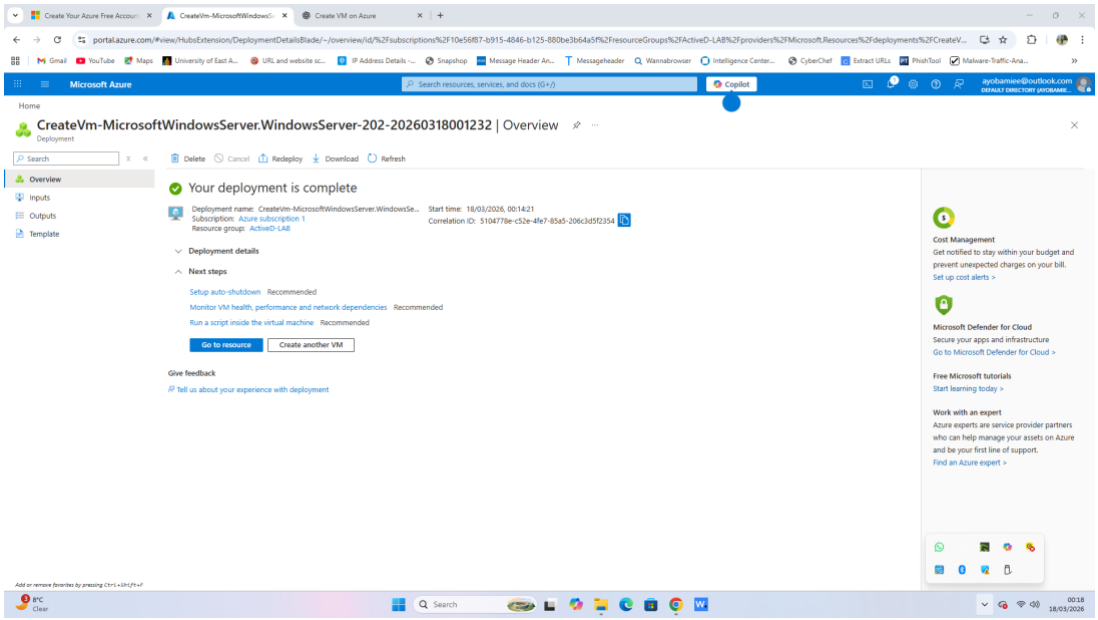


- Create a Virtual network named **(ActiveD-Vnet)** and it was deployed to the resource group.



- **Provisioning Cloud Virtual Machines:**

  1. Create VM 1 named **(Domain-Controller01).** This VM is used to host the active directory domain services (AD DS), DNS and serve as the primary domain controller of the environment. It is also deployed under the same resource group.
  2. Configuration:
     a. OS (Window server 2022)
     b. Size: B2als_v2 (It is cost efficient)
     c. Network: ActiveD-VNet
     d. Subnet: default
     e. Public IP: Optional (recommended only for RDP access
     f. Private IP: Static (recommended for domain controllers)
     g. Inbound Ports: RDP (3389)

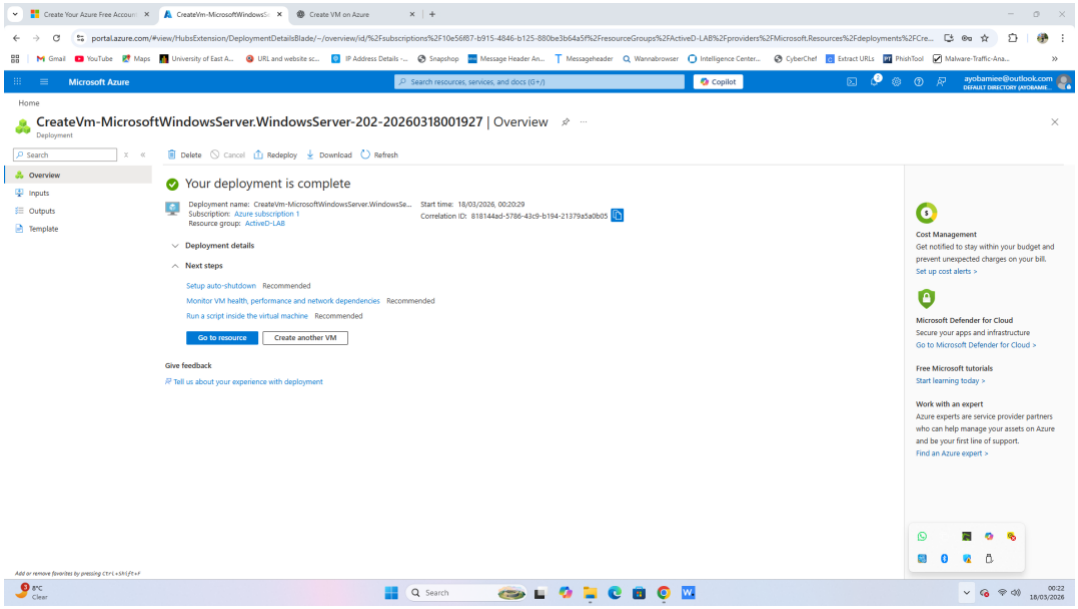


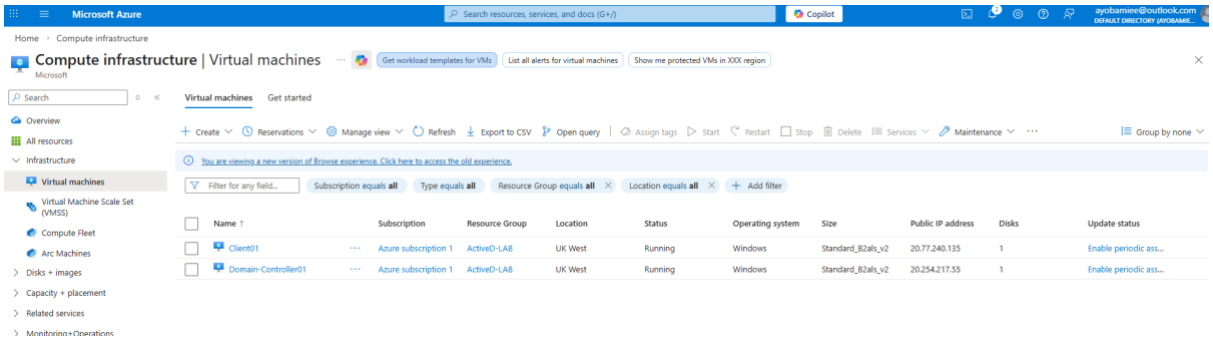
Create VM2 named **(Client01).** This VM acts as a domain-joined workstation used to test authentication, group policies and user account management.

Configuration:

a. OS: Window server 2022
b. Size: B2als_v2
c. Network: ActiveD-VNet
d. Subnet: default
e. Private IP: Dynamic is acceptable
f. Inbound Ports: RDP (3389)
g. DNS Settings: Must point to Domain-Controller01's private IP before joining the domain

Notes: Keep OS patched before domain join for secure configuration.



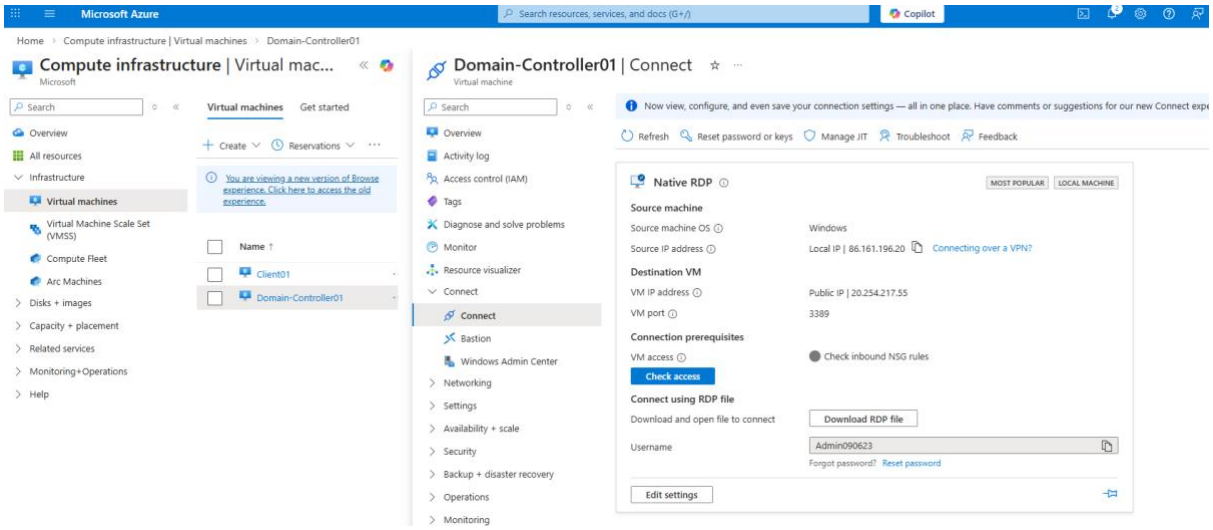


Note: The VMs must be in the same virtual network (ActiveD-VNet) and the same region (UK-west). This allows:

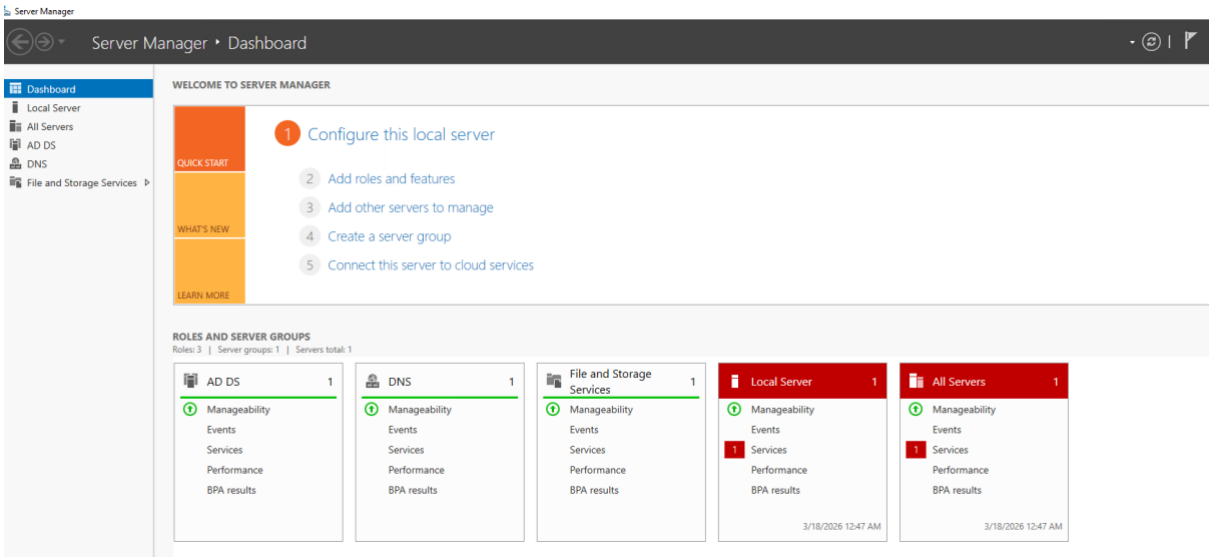
- Domain join
- DNS resolution
- Group Policy communication
- Realistic enterprise LAN simulation

Configuring the VM(Domain-Controller01)

- Go to Virtual Machines → select **Domain-Controller01** → click Connect.
- Download the RDP and open it



- Use the credentials you set when creating the VM.
- The Remote Desktop session will open the Windows Server desktop.
- Open Server Manager (usually starts automatically at login)



- Click Add Roles and Features.
- In the Add Roles and Features Wizard:
- Installation Type: Select Role-based or feature-based installation → Next.
- Server Selection: Select **Domain-Controller01** → Next.
- Server Roles: Check Active Directory Domain Services → a pop-up will ask to add features → click Add Features → Next.

- Features: Leave default → Next.
- AD DS: Read description → Next.
- Confirmation: Check “Restart the destination server automatically if required” → click Install.

After installation, click the Notifications flag in Server Manager → Promote this server to a domain controller.

Deployment Configuration:

- Choose Add a new forest → Root domain name: corp.local → Next.

Domain Controller Options:

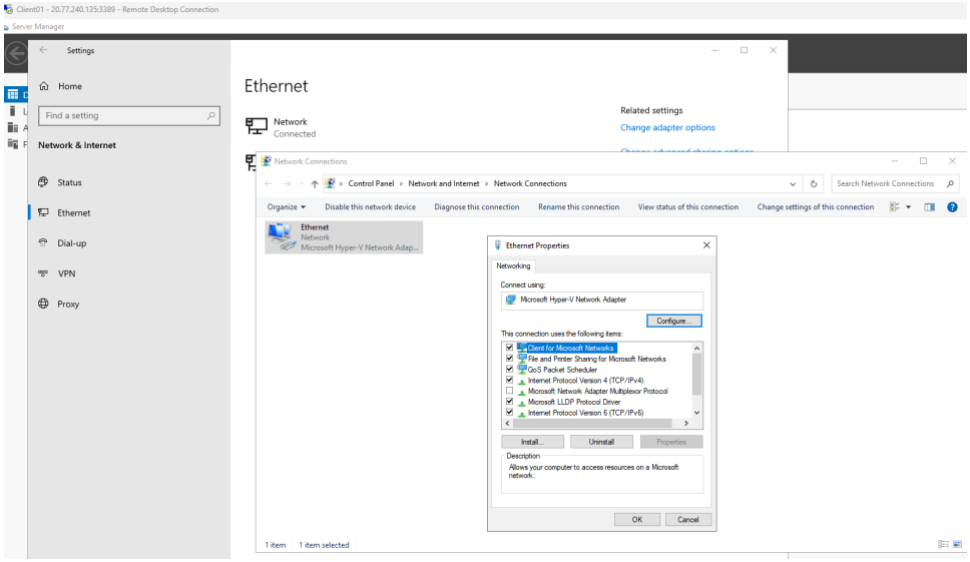
- Forest and Domain Functional Levels: Default (Windows Server 2022 or latest version).
- Check the DNS Server → Next.
- Set a DSRM password (Directory Services Restore Mode).
- Paths: Leave default (C:\Windows\NTDS, etc.) → Next.
- Review Options: Confirm → Next.
- Click Install → The server will automatically reboot after promotion.
-

After reboot, **Domain_controller01** is now the Domain Controller for the new forest **corp.local.**

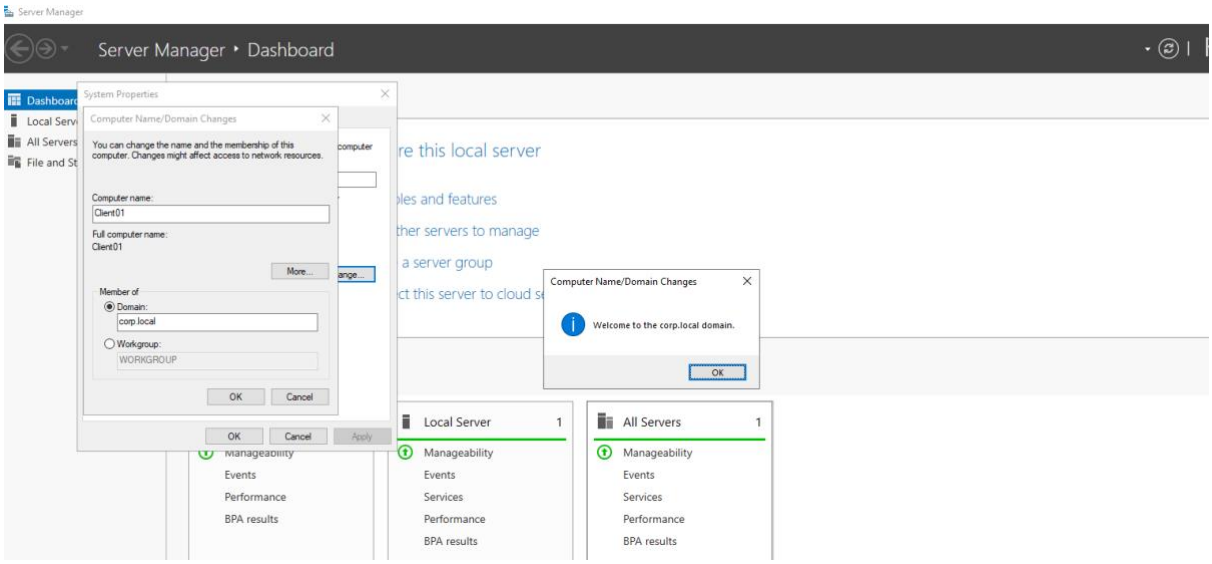
Note: During the domain setup, a domain administrator account named ***"admin090623"*** was created and has Domain Admin privileges and used for administrative tasks in this project.

Configuring the endpoint (Client01) VM to the domain

- Go to Virtual Machines → select **Client01** → click Connect.
- Download the RDP and open it
- Log on to the VM with the credentials from the VM set up
- Open Settings → Network & Internet → Ethernet → Change adapter options →
- Right-click your adapter → Properties → IPv4 → Properties → Preferred DNS
- server → Enter Domain-Controller01 private IP → OK.



- Open System Properties: Right-click This PC → Properties → Advanced system settings →Computer Name tab → Change…



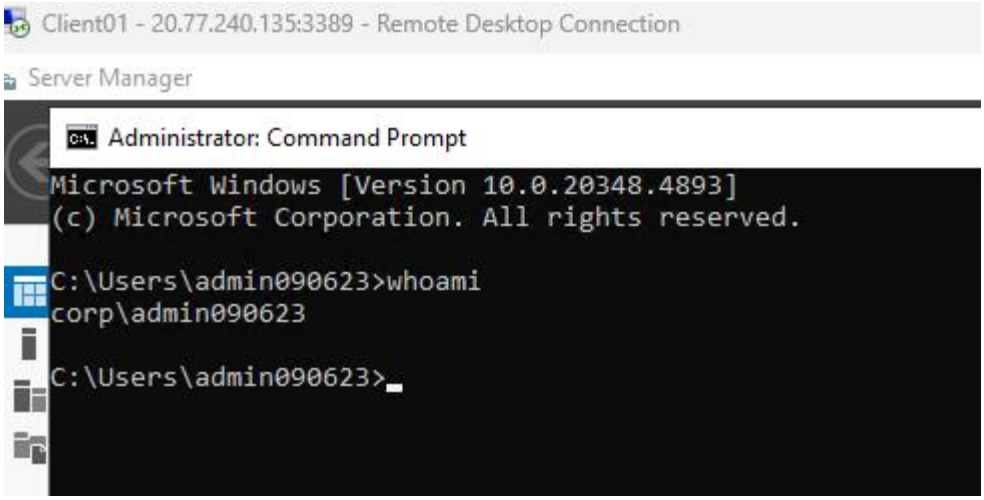
Select Domain

- Enter the domain: corp.local –>Click OK.
- Enter Domain Admin credentials (admin090623) from Domain-Controller01.
- Restart Client01 to apply domain membership.

After reboot, the workstation is now domain-joined.

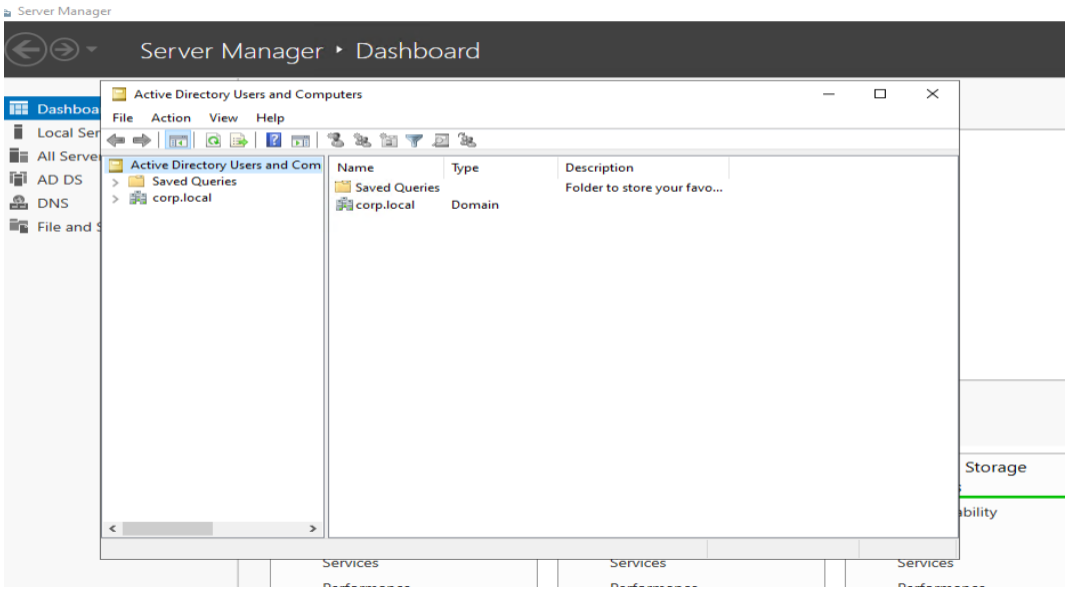
To Verify the Domain Membership in the Client01 VM:

- Log in with domain admin credentials (admin090623) in the client01 VM
- After log-on, Check the current account using the Command Prompt: *Whoami*

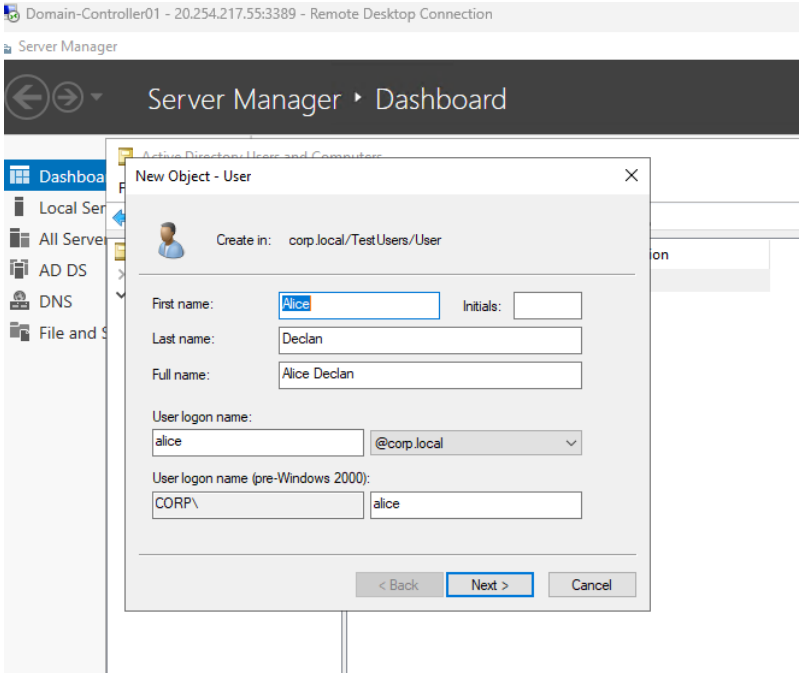


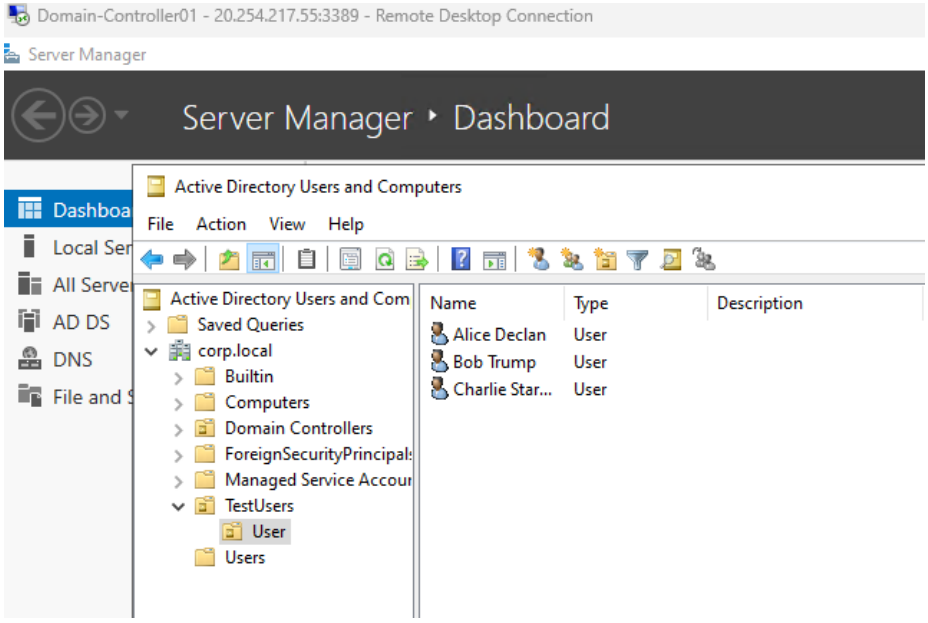
Create Users and Organizational Units

- On VM1 (Domain-Controller01): Server Manager → Tools → Active Directory Users and Computers (ADUC).
- Right-click your domain → New → Create an Organizational Unit (TestUsers)



- Inside the Organizational unit (TestUsers), Create another folder named (Users)
- Create users: Alice Declan, Bob Trump & Charlie
- Set password → Require password change at next login
- Ensure all users are added to the Domain Users group.


Domain-Controller01 - 20.254.217.55:3389 - Remote Desktop Connection
Server Manager
Server Manager ‣ Dashboard
Dashboa
Local Ser
All Server
AD DS
DNS
File and S
New Object - User
Create in: corp.local/TestUsers/User
First name: Alice
Initials:
Last name: Declan
Full name: Alice Declan
User logon name:
alice
@corp.local
User logon name (pre-Windows 2000):
CORP\
alice
< Back
Next >
Cancel


Domain-Controller01 - 20.254.217.55:3389 - Remote Desktop Connection
Server Manager
Server Manager ‣ Dashboard
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Local Ser
All Server
AD DS
DNS
File and S
Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
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Builtin
Computers
Domain Controllers
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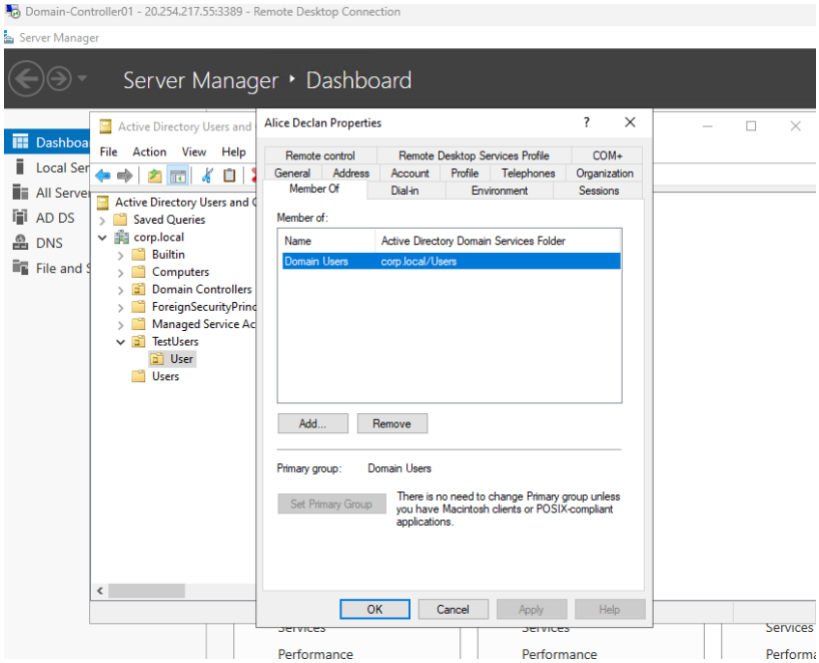
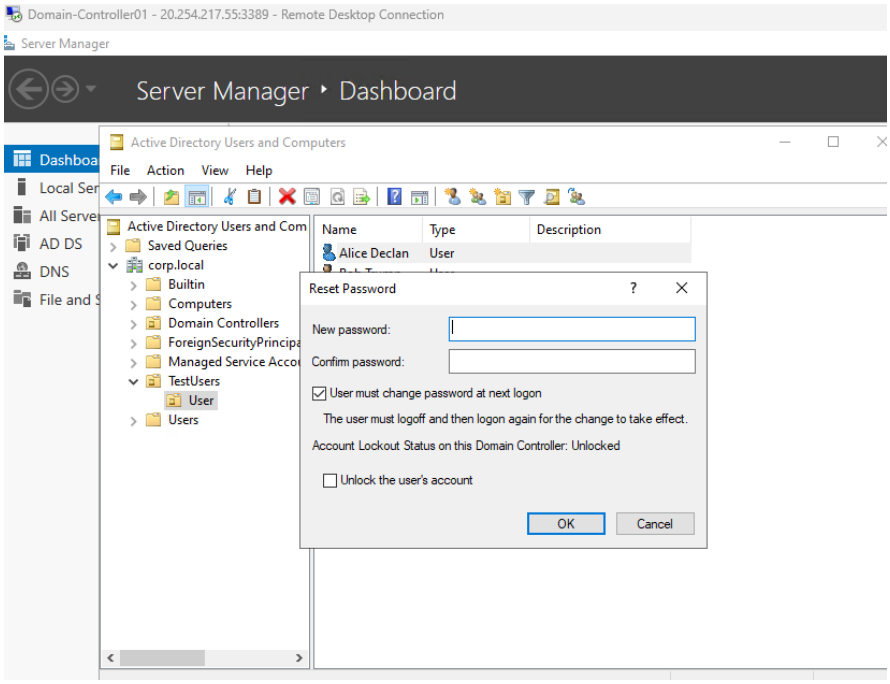
TestUsers
User
Users
Name
Type
Description
Alice Declan
User
Bob Trump
User
Charlie Star...
User



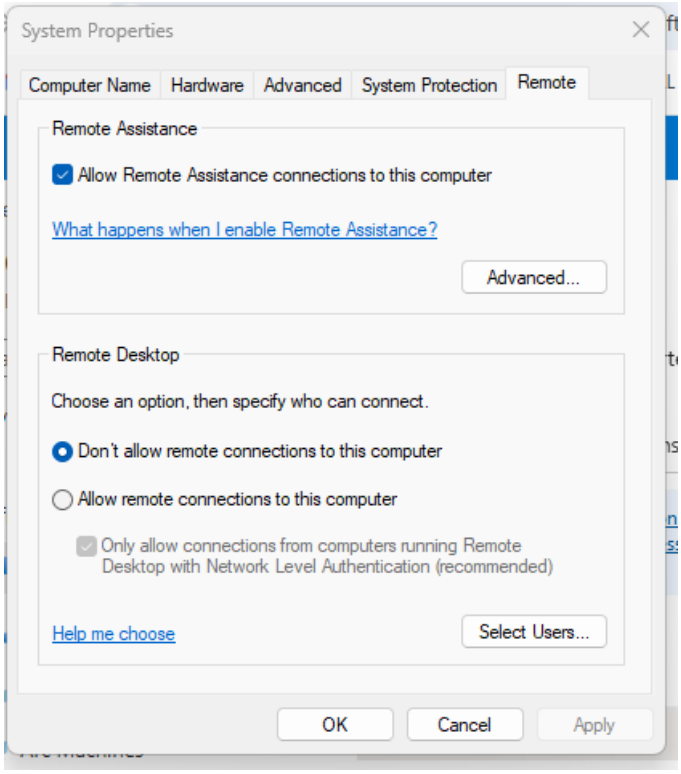
This simulates enterprise identity lifecycle management, a key focus area for SOC monitoring and audit investigations.

Reset Password & Add User to “Remote Desktop Users” Group

On the domain-controller01, you can reset the client password and add the client to the remote desktop user group.

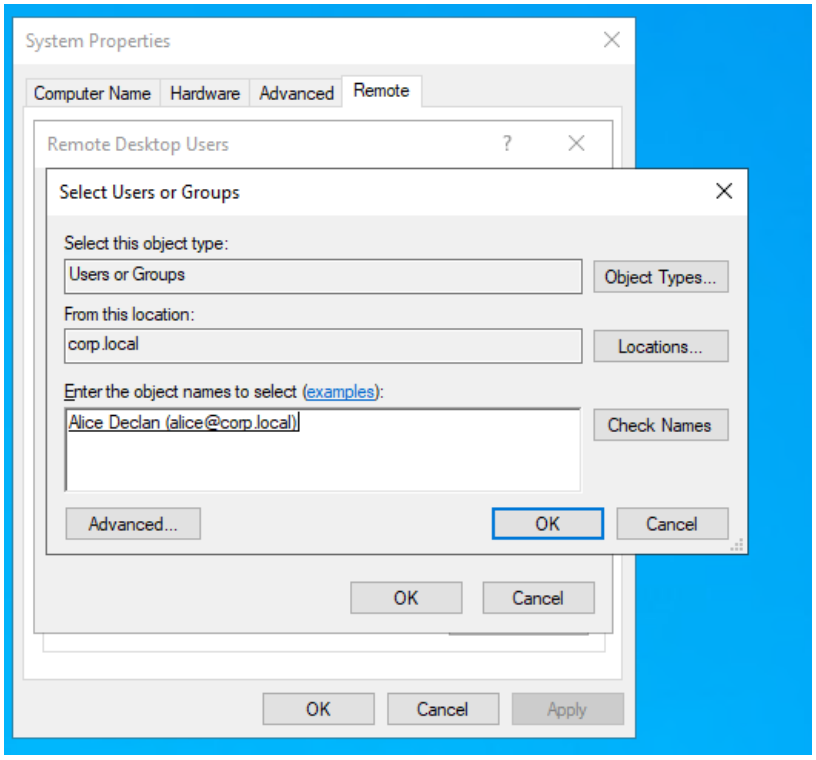


This fixes the issue with changing the password at next logon.



This adds Alice to the Remote Desktop Users local group.

- On the **Client01** machine, you can RDP into the machine using the domain admin credentials**(admin090623)**
- Open the system properties: Win + R –> sysdm.cpl
- Go to the remote tab –> Add user –> Corp\Alice

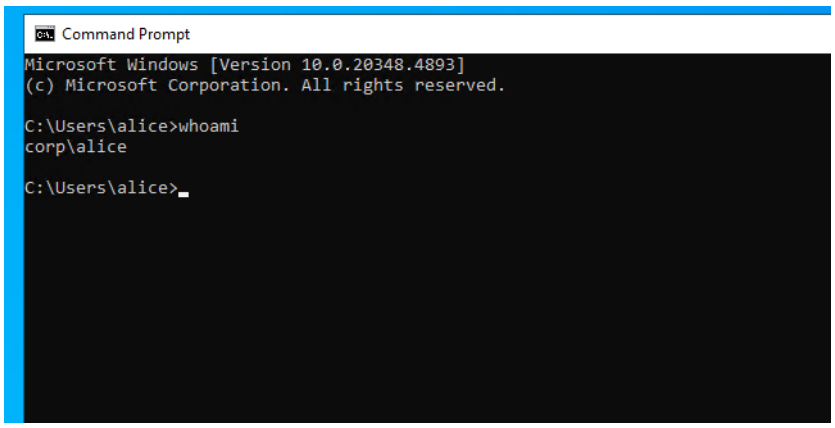


Confirm the Remote Desktop User (Alice) can log on into the Client01(VM2).

- To Ensure users can log in and communicate with domain resources.
- On Client01 (VM2)→ log in with a new domain account (corp\alice)
- Enter the credentials

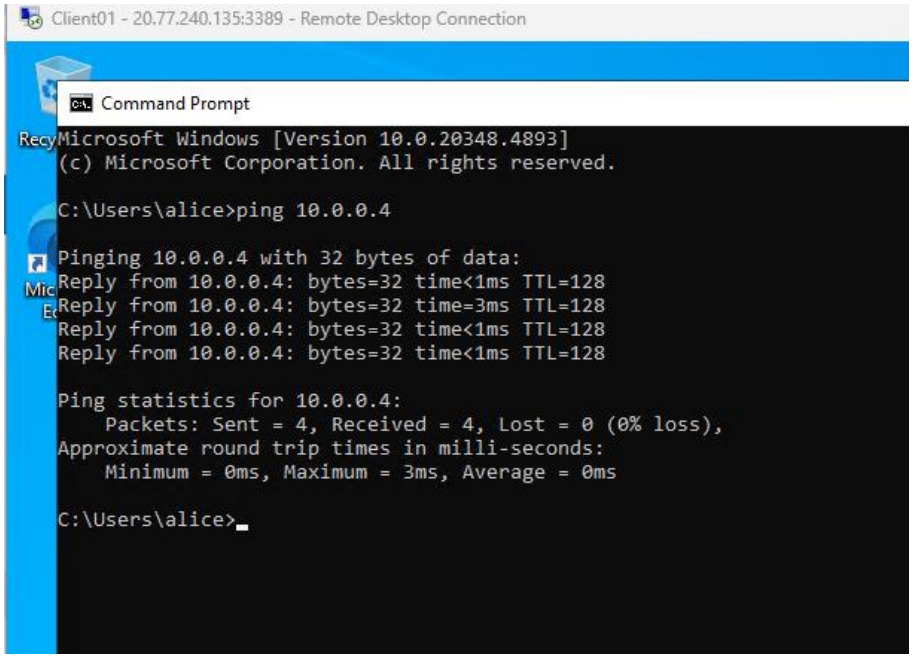
Verify Domain Access.

- Run the *whoami* command prompt



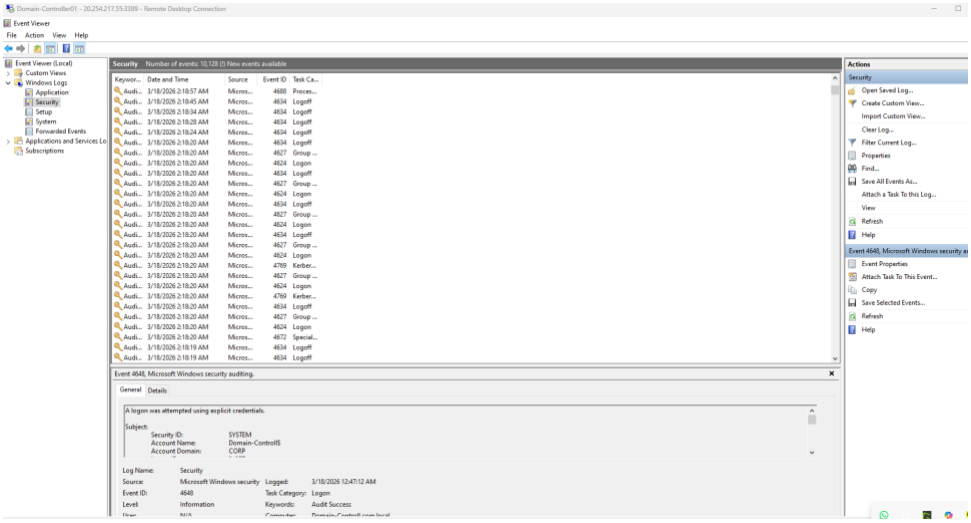
Test network Access:

- Ping the domain-controller01 private IP address.



Window Security Log Monitoring on Event Viewer

- On Domain-Controller01, Access the Event viewer–>Window logs–>Security.



This log contains authentication events, account management activity, and policy changes, which are critical for SOC investigations.

Identify Key Active Directory Security Event ID

Authentication Event ID

Event ID	Description	SOC relevance
4624	Successful Logon	To confirm user authentication.
4625	Failed Logon	To detect brute-force or password spraying attack.
4648	Login using explicit credentials	Indicates lateral movement attempts.

Account management Event ID

Event ID	Description	SOC relevance
4720	User Account Created	Track User Provisioning
4722	User Account Enabled	Detected Reactivated Account.
4723/4724	Password Change/Reset	Monitor credentials changes/resets
4725	User Account Disabled	Track Access Revocation
4726	User Account Deleted	Identifies Account removal

Domain and Group Changes Event ID

Event ID	Description	SOC relevance
4732	User added to a security group	Monitoring privileged escalation
4733	User removed from a security group	Track access changes
4735	Security group modified	Detects group tampering

Note: Some windows security event IDs may vary depending on the window server version, client OS version, domain functional level or Audit policy configuration.

SOC Analysts can export the logs from the event viewer and parse into a SIEM(Splunk, wazuh or Sentinel) for centralized analysis and alerting.

CONCLUSION

This project successfully demonstrates the deployment and administration of a cloud-based Active Directory environment within Microsoft Azure, simulating a real-world enterprise infrastructure. Through the configuration of domain services, user management, and endpoint integration, the project establishes a foundational understanding of identity and access management.

From a security operations perspective, the project highlights the importance of monitoring authentication events, account management activities, and group membership changes using Windows Security Event Logs. These logs form a critical component of SOC workflows, enabling analysts to detect suspicious behaviours such as brute-force attempts, unauthorized privilege escalation, and potential lateral movement.

While the current implementation focuses on infrastructure and log visibility, future enhancements could include SIEM integration, automated alerting, and attack simulation scenarios to strengthen detection and response capabilities.